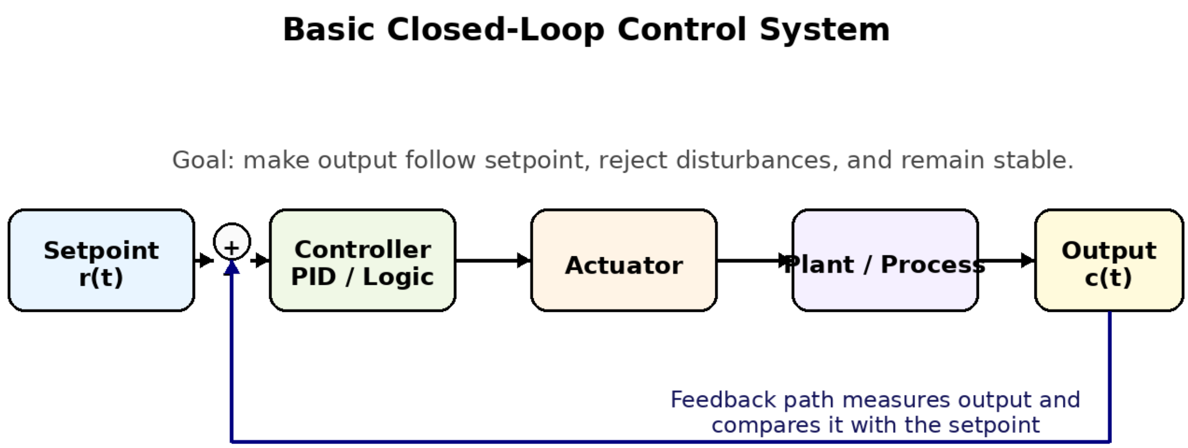
Industrial Automation & Control - Consolidated Notes

Weeks 2, 3, 6 and 7 | Detailed but exam-focused | Separate notes volume



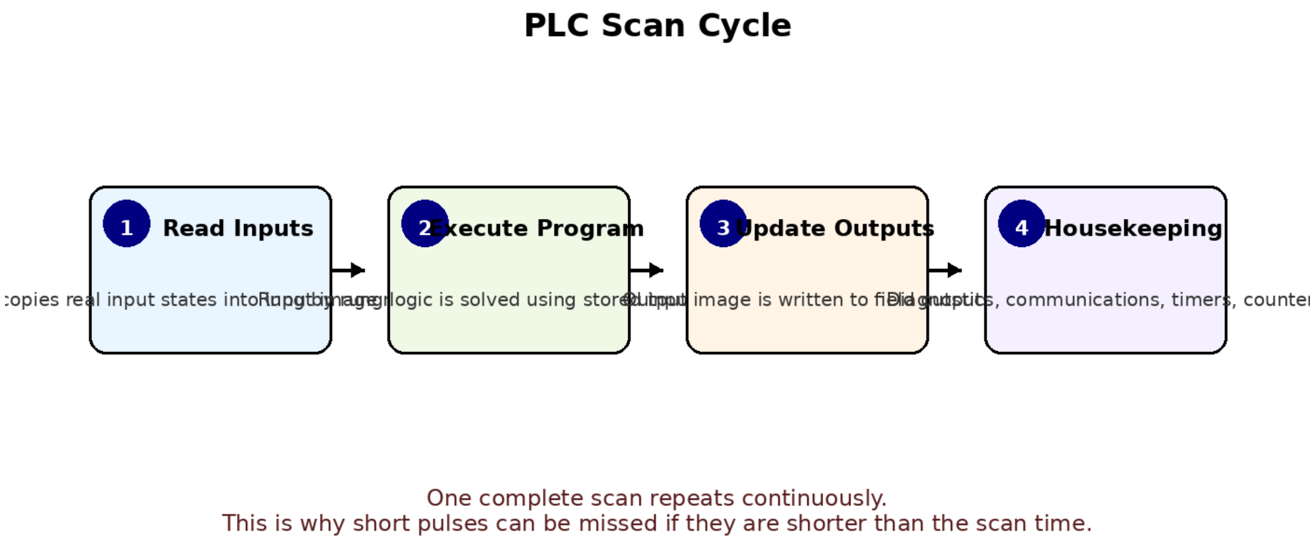
Prepared from the uploaded NPTEL assignment PDFs for Weeks 2, 3, 6 and 7.
Structured for exam revision with explanations from scratch.

How to use this notes volume

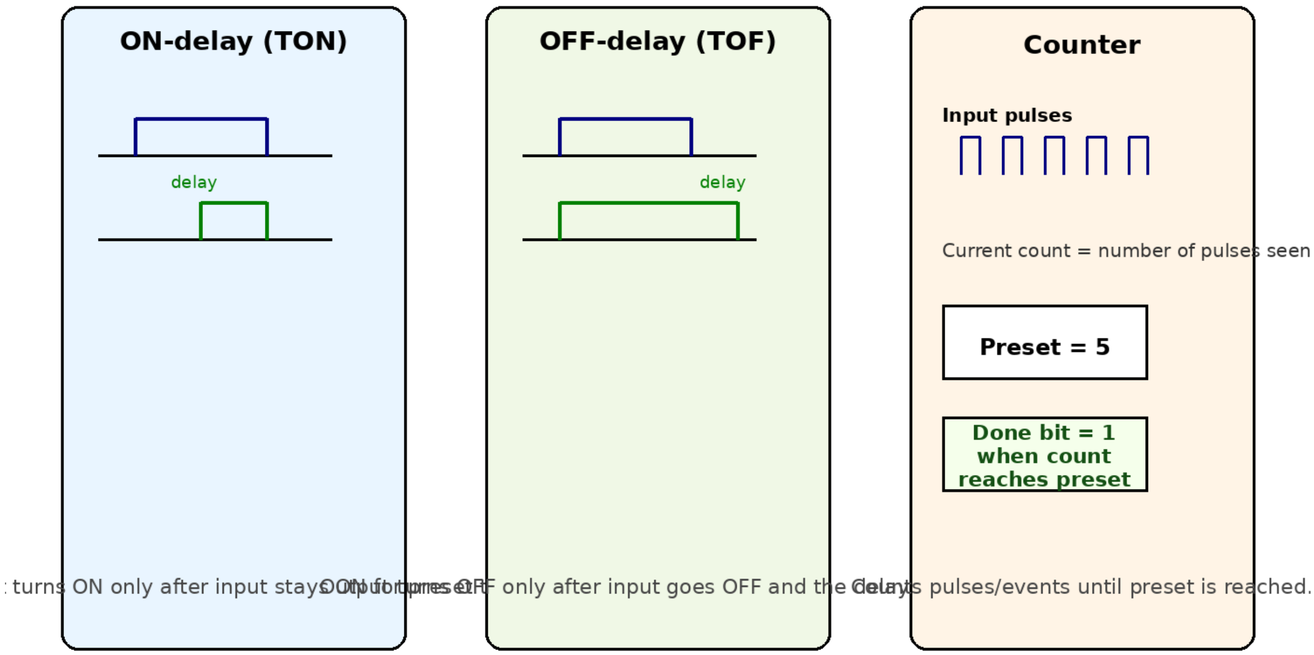
• Start with the section banners and memorise the highlighted definitions.

• For numericals, remember the formulas and the logic behind them, not just the final number.

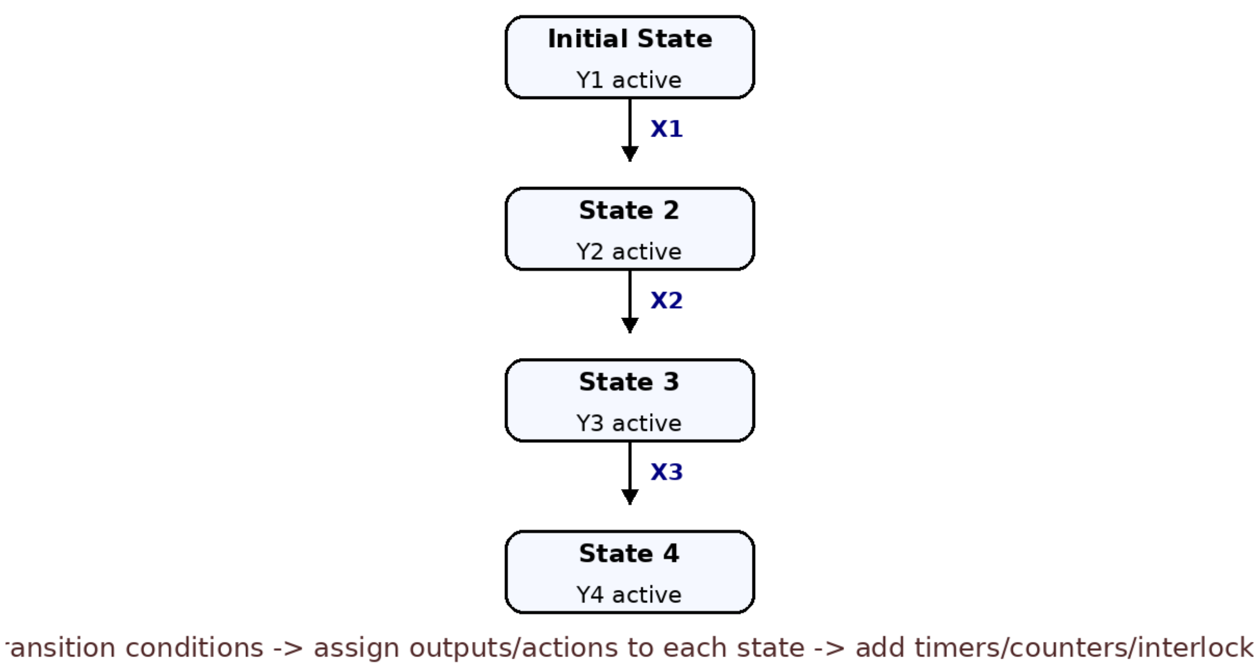
• For PLC/sequence questions, always convert words -> logic condition -> states/steps -> final ladder.



Timer and Counter Intuition



From Description -> FSM / SFC -> Ladder Logic



Weeks 2 - Measurement & Data Acquisition

Static characteristics

• Accuracy = closeness to the true value. Precision = closeness among repeated readings.

• Sensitivity is the slope of the input-output curve: change in output per unit change in input.

• Resolution is the smallest change that can be detected.

• Dead zone means small input changes produce no output change.

• Hysteresis means output depends on whether input is increasing or decreasing.

• Drift is unwanted change with time even when input is constant; includes zero drift and span drift.

• Linearity tells how closely the calibration curve matches a straight line.

• Error may be absolute, relative, or percentage.

Dynamic characteristics

• Zero-order element: practically instantaneous, no storage, example potentiometer.

• First-order element: one storage element; time constant τ controls speed; at $t = \tau$, output reaches 63.2% of final value for a step input.

• Second-order element: two storage elements; damping ratio ζ determines underdamped / critically damped / overdamped behavior.

• Overshoot occurs in underdamped systems. More damping reduces overshoot but may slow response.

• Resonance occurs near natural frequency in lightly damped systems.

Data acquisition systems

• Typical path: physical process -> sensor -> signal conditioning -> sample and hold -> ADC -> computer/display.

• Signal conditioning may include amplification, isolation, filtering, and linearization.

• Anti-aliasing filter is a low-pass filter placed before the ADC.

• Nyquist rule: sample faster than twice the highest signal frequency.

• Sample-and-hold freezes the analog value during ADC conversion.

• Flash ADC is fastest but becomes impractical at large bit counts.

• SAR ADC is widely used and usually needs about n clock cycles for n bits.

• R-2R ladder DAC uses only two resistor values, R and 2R, which simplifies matching.

Memory hook: accuracy = closeness to truth, precision = closeness to each other, sensitivity = slope, resolution = smallest detectable change.

Weeks 3 - Automatic Control & PID

Automatic control basics

• Open-loop control does not compare output with reference. Closed-loop control uses feedback.

• Main goals: stability, setpoint tracking, disturbance rejection, low steady-state error, and good transient behavior.

• Steady-state performance describes final error after transients die out.

• Transient performance describes rise time, overshoot, settling time, and oscillation.

Stability and phase lag

• A stable system settles to a bounded value after a bounded input or disturbance.

• Phase lag comes from delays and dynamic elements in sensors, actuators, and the plant.

• Too much loop gain combined with phase lag can destabilize feedback.

PID actions from scratch

• Proportional action gives output proportional to error. It speeds response but alone usually cannot eliminate steady-state error.

• Integral action accumulates error over time and drives steady-state error toward zero. Too much integral can cause oscillation and windup.

• Derivative action reacts to the rate of change of error. It improves damping and reduces overshoot, but it is sensitive to noise.

• Practical PID controllers often filter or limit derivative action.

• When actuators saturate, integral windup can occur. Anti-windup logic is used to prevent large stored integral action.

PID forms and tuning intuition

• Position form computes full controller output u(k). Velocity form computes change in output Δu(k).

• Proportional band is inversely related to proportional gain.

• Increasing Kp usually reduces rise time but can increase overshoot.

• Increasing integral action reduces steady-state error but can worsen oscillation.

• Increasing derivative action usually reduces overshoot and improves damping.

• Bumpless auto/manual transfer prevents sudden jumps when switching modes.

Exam hook: P improves speed, I removes offset, D improves damping.

Weeks 6 & 7 - PLC, RLL, Timers, Counters, Structured Sequence Design

PLC and relay ladder logic basics

• PLC = programmable logic controller used for industrial automation.

• RLL (relay ladder logic) represents logic with contacts and coils. Series = AND, parallel = OR, normally closed contact = NOT.

• Auxiliary contacts let an output status be reused inside other rungs.

• Interlocks prevent unsafe or conflicting outputs from turning on together.

• Emergency-stop paths are designed fail-safe, usually using normally closed hardware contacts.

Scan cycle

• During each PLC scan, inputs are read, logic is solved, outputs are updated, and housekeeping/communications run.

• Because logic uses an input image during a scan, very short pulses can be missed unless captured by special hardware or fast tasks.

Timers and counters

• TON (on-delay): output becomes true only after input remains true for preset time.

• TOF (off-delay): output remains true for preset time after input goes false.

• Pulse timer: creates a fixed-width pulse when triggered.

• Counters count events/pulses until preset is reached. A done bit becomes true at the preset count.

• Reset logic clears timer/counter state when the sequence requires restarting.

Structured design approach

• Never jump directly from a word problem to random ladder rungs. First identify inputs, outputs, states, modes, and safety conditions.

• Convert the description into a state diagram, finite-state machine (FSM), or sequential function chart (SFC).

• Define actions associated with each state, then define transition conditions.

• Only after the model is clear should you translate it into ladder logic.

• Large systems are memory-based: the same input may need different outputs depending on the current state.

• Formal models are essential for reliable automatic code generation.

SFC and parallel branches

• An initial step is active after power-on or reset.

• Alternative transitions use priority if more than one becomes true together; commonly the leftmost transition wins.

• Parallel branches are legal when properly synchronized, but jumping into or out of a parallel sequence is illegal.

• State outputs remain associated with the active step until transition conditions move execution onward.

Industrial communication quick note

• RS-232 is short-distance, point-to-point, and more noise-sensitive.

• RS-485 supports longer distance, multiple devices, and better noise immunity through differential signaling.

• In process/sequence control systems, hardware, software, control algorithm, and plant dynamics must all be considered together.

Exam hook: series = AND, parallel = OR, NC = NOT, latch = output contact in parallel with start contact.

Last-minute 1-page recall list

• Zero-order -> no storage; First-order -> time constant; Second-order -> damping ratio and overshoot.

• Nyquist: $f_s > 2f_{max}$. Anti-aliasing filter is low-pass.

• Step steady-state error for type-0 unity feedback: $e_{ss} = 1/(1+K_p)$.

• PB is inversely proportional to Kp.

• Saturation can make feedback behave effectively open-loop.

• TON delays turn-on. TOF delays turn-off. Counter done bit turns on at preset count.

• In sequence design: define states before coding ladder.